

1.2 Memory and storage

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.2 Memory and storage
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture

After it: 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Memory bay ms-101	What are the differences between RAM and ROM?	11	27
2	Swap space ms-102	Why do we need virtual memory, and what can we do to avoid using it?	12	20
3	Storage warehouse ms-103	Why do computers have secondary storage?	12	25
4	Top Trumps ms-104	What are the differences between secondary storage devices?	12	24
5	Pick the storage ms-105	What features of secondary storage make devices suitable for different situations?	12	23
6	Units lab ms-106	Why is data stored in binary?	15	27
7	Capacity calculator ms-107	How do you calculate data capacity?	15	15

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
8	Binary workshop ms-108	What can happen when you add two binary numbers together?	15	15
9	Assessment 1 ms-109-test	assesses the whole unit	0	0
10	Shift and hex lab ms-110	What is the relationship between denary, binary and hexadecimal?	15	15
11	Revision HQ 1.2 part 1 ms-111	stretch work, no new content	21	21
12	Character foundry ms-112	How does a computer store characters, and what are the implications for the number of bits used?	16	23
13	Pixel studio ms-113	How does a computer store graphics?	15	15
14	Sound studio ms-114	How do computers store sound, and what are the implications for sample rate, duration and bit depth?	15	15
15	Compression works ms-115	What are the effects on a file for each type of compression?	14	24
16	Revision HQ 1.2 part 2 ms-116	stretch work, no new content	20	27
17	Assessment 2 ms-117-test	assesses the whole unit	0	0
18	Binary blitz ms-bonus	stretch work, no new content	1	0

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1

Vocabulary introduced: RAM, ROM, Virtual memory, Volatile, Secondary storage, Non-volatile, Capacity, Durability, Portability, Suitability, Bit, Byte, Data capacity, File size, Binary, Overflow error, Binary shift, Hexadecimal, Storage, Units, Hex, Character set, Unicode, Pixel, Metadata, Sampling, Sample rate, Lossy compression, Lossless compression, ASCII, Pixels, Compression

What a pupil does, not only reads: addshift, blitz, code, convert, match, mcq, multi, order, pixels, sort, sound

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand the need for primary storage ms-101-o1	ms-l01	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know the difference between RAM and ROM ms-l01-o2	ms-l01	yes	yes	yes	Full
know the purpose of RAM and ROM ms-l01-o3	ms-l01	yes	yes	yes	Full
understand the need for virtual memory ms-l02-o1	ms-l02	yes	yes	yes	Full
understand how virtual memory works ms-l02-o2	ms-l02	yes	yes	yes	Full
understand the need for secondary storage ms-l03-o1	ms-l03	yes	yes	yes	Full
know the common types of storage ms-l03-o2	ms-l03	—	—	yes	Partial
know how each type works ms-l03-o3	ms-l03	—	—	yes	Partial
know the characteristics of storage devices ms-l04-o1	ms-l04	yes	yes	yes	Full
be able to compare devices using those characteristics ms-l04-o2	ms-l04	yes	yes	yes	Full
understand the suitability of storage devices for certain applications ms-l05-o1	ms-l05	—	—	yes	Partial
understand the advantages and disadvantages of devices based on their characteristics ms-l05-o2	ms-l05	—	—	yes	Partial
know the units bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte and petabyte ms-l06-o1	ms-l06	yes	yes	yes	Full
be able to convert between units ms-l06-o2	ms-l06	yes	yes	yes	Full
understand that data must be converted to binary to be processed ms-l06-o3	ms-l06	—	—	yes	Partial
know what data capacity means ms-l07-o1	ms-l07	yes	yes	yes	Full
understand how to calculate data capacity requirements ms-l07-o2	ms-l07	yes	yes	yes	Full
convert positive denary whole numbers (0-255) to 8-bit binary and back ms-l08-o1	ms-l08	yes	yes	yes	Full
add two 8-bit binary integers ms-l08-o2	ms-l08	yes	yes	yes	Full
understand the terms most and least significant bit ms-l08-o3	ms-l08	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand how overflow errors occur ms-l08-o4	ms-l08	yes	yes	yes	Full
know how to perform left and right binary shifts ms-l10-o1	ms-l10	yes	yes	yes	Full
understand what a binary shift achieves ms-l10-o2	ms-l10	—	—	yes	Partial
convert denary to 2-digit hexadecimal and back ms-l10-o3	ms-l10	yes	yes	yes	Full
convert between binary and hexadecimal ms-l10-o4	ms-l10	yes	yes	yes	Full
know which parts of the topic I am secure on ms-l11-o1	ms-l11	—	—	—	Not covered
finish any outstanding tasks from lessons 1 to 10 ms-l11-o2	ms-l11	—	—	—	Not covered
understand that all data, including text, is represented in binary ms-l12-o1	ms-l12	yes	yes	yes	Full
know what is meant by a character set ms-l12-o2	ms-l12	yes	yes	yes	Full
understand the relationship between the number of bits and the number of characters ms-l12-o3	ms-l12	yes	yes	yes	Full
know about ASCII and Unicode ms-l12-o4	ms-l12	yes	yes	yes	Full
understand how an image is a series of pixels represented in binary ms-l13-o1	ms-l13	yes	yes	yes	Full
know what is meant by metadata ms-l13-o2	ms-l13	yes	yes	yes	Full
understand the effect of colour depth and resolution on file size ms-l13-o3	ms-l13	yes	yes	yes	Full
understand how sound is sampled and stored in digital form ms-l14-o1	ms-l14	yes	yes	yes	Full
understand how sample rate, duration and bit depth affect file size and quality ms-l14-o2	ms-l14	yes	yes	yes	Full
know why data is compressed ms-l15-o1	ms-l15	yes	yes	yes	Full
understand the difference between lossy and lossless compression ms-l15-o2	ms-l15	yes	yes	yes	Full
know why some data can only be compressed losslessly ms-l15-o3	ms-l15	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know which parts of data representation I am secure on ms-l16-o1	ms-l16	yes	yes	yes	Full
finish outstanding tasks from lessons 12 to 15 ms-l16-o2	ms-l16	—	—	—	Not covered

32 full · **6 partial** · **3 not covered**, of 41 outcomes.

Gaps, assumptions and overlap

- **ms-l03-o2** know the common types of storage — no station's text matches it
- **ms-l03-o3** know how each type works — no station's text matches it
- **ms-l05-o1** understand the suitability of storage devices for certain applications — no station's text matches it
- **ms-l05-o2** understand the advantages and disadvantages of devices based on their characteristics — no station's text matches it
- **ms-l06-o3** understand that data must be converted to binary to be processed — no station's text matches it
- **ms-l10-o2** understand what a binary shift achieves — no station's text matches it

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- ms-l03-x2 Describe how a hard disk drive stores data.
- ms-l03-x3 Describe how data is read from an optical disc.
- ms-l04-x2 Explain why an SSD is more suitable than a hard disk for a laptop.
- ms-l05-x2 Explain why a school uses magnetic hard disks rather than SSDs for its main file storage.
- ms-l06-x3 A file is 2,500 MB. Convert this to gigabytes.
- ms-l09-test-x4 Characteristics and suitability
- ms-l09-test-x5 Units of storage
- ms-l09-test-x6 Capacity calculations
- ms-l09-test-x7 Binary conversion and addition
- ms-l17-test-x3 Character sets

What could not be traced

The wall text of ms-bonus is painted into the 360 image and held nowhere as text, so an outcome in that lesson can only be traced through its activities and its exam questions. Read the rows above for it with that in mind.

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